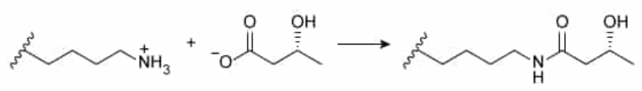
In addition to being metabolized for energy, the ketone body β-hydroxybutyrate (BHB) can also react with lysine residues of histone proteins.

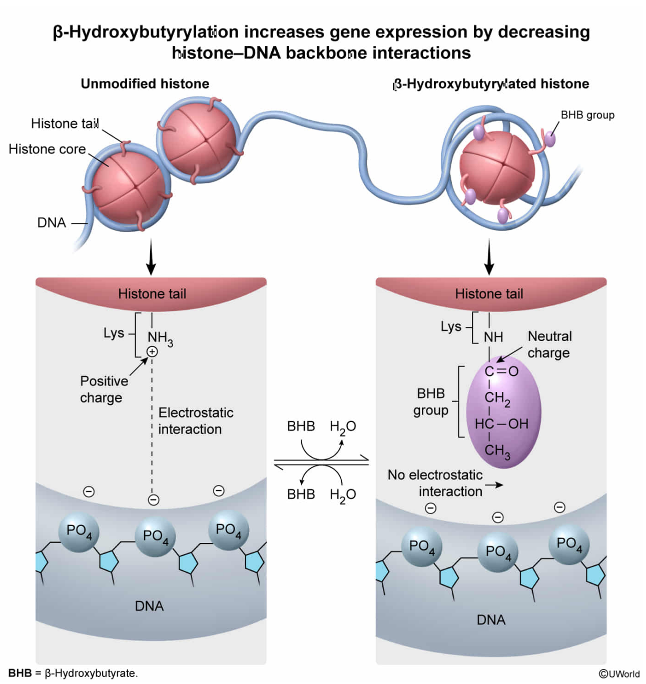


Which statement best explains the general upregulation of gene expression following histone β-hydroxybutyrylation? β-Hydroxybutyrylation:

- ✔ A. decreases the binding interaction between histones and the DNA backbone. [63%]
- B. increases the binding interaction between histones and the nitrogenous bases of DNA. [25%]
- C. increases the binding interaction between histones and the RNA backbone. [5%]
- D. decreases the binding interaction between histones and the nitrogenous bases of RNA. [4%]

Correct
64% answered correctly

Explanation:



Histones are nuclear proteins that help in the organization of eukaryotic DNA. Histones serve as structural support for DNA to wind tightly around, allowing genomic DNA to be condensed into the nucleus of each cell. Histones can also help regulate gene expression, depending on the **strength of the binding**

interaction between the histone and its associated DNA—strong interactions inhibit the DNA's accessibility and therefore prevent transcription and gene expression whereas **weaker interactions promote transcription and gene expression**.

The question states that β-hydroxybutyrylation modifies the lysine residues of histones. Lysine residues are **positively charged** and play an important role in histones due to their **electrostatic interaction** with the **negatively charged phosphate** groups of DNA's sugar-phosphate backbone. Reaction with β-hydroxybutyrate (BHB) **neutralizes** the lysine side chain, replacing the positively charged amine with a **neutral and non-ionizable** amide. By neutralizing the positive charge, β-hydroxybutyrylation of lysine causes the **DNA–histone interaction to weaken**, which leads to upregulation of gene expression.

(Choice B) Histones interact with the *backbone* of DNA, not with the nitrogenous bases. Furthermore, a strengthened interaction between DNA and histones is likely to *decrease* the DNA's accessibility and *downregulate* gene expression, not upregulate it.

(Choices C and D) Although RNA is the product of upregulated gene transcription, generally RNA does not interact with histones. Histones interact with *DNA*, not RNA.

Educational objective:
Histones are proteins that bind DNA through electrostatic interactions between the negatively charged DNA backbone and positively charged histone residues like lysine. Modifications that remove the positive charge loosen the DNA–histone interaction and lead to an upregulation of gene expression.

Time spent:0 sec
Subject:
Biochemistry

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Amino Acids and Proteins